
SKILLS

- **Quant Development:** Python, Go, SQL, research tooling, backtesting, market data streams, portfolio analytics, risk controls
- **Data Engineering:** Kafka, Airflow, Snowflake, PostgreSQL, Redis, Elasticsearch, ETL, schema design, time-series analytics
- **Quant Analytics:** Monte Carlo simulation, Sharpe analysis, anomaly detection, portfolio optimization
- **Trading Systems:** market data ingestion, momentum/mean-reversion signals, position sizing, risk controls
- **Software Design:** API design, microservices, reusable components, design patterns, requirements analysis
- **Engineering Process:** testing strategy, code reviews, agile delivery, technical specs, project ownership
- **Production Systems:** CI/CD, Git, Docker, Kubernetes, Terraform, Linux, logging, monitoring, alerting, level-2 support
- **Observability:** OpenTelemetry, ELK, CloudWatch, distributed tracing, SLOs, incident response, production debugging
- **Collaboration:** ADRs, runbooks, stakeholder demos, researcher workflows, cross-team platform standards

EXPERIENCE

- **Royal Bank of Canada (RBC)** Toronto, Canada
Staff Software Developer & Enterprise Architect – Data / Platform Nov 2022 – Present
 - **High-Scale Data & Observability Platform:** Architected institution-wide telemetry platform using OpenTelemetry, Vector, Fluent Bit, Kafka, and Logstash ingesting 150 TB/day across 10,000+ systems; standardised schemas for time-series analytics, alerting, SLOs, anomaly detection, and production debugging.
 - **Reusable Research-Style Tooling:** Led 0-to-1 delivery of patent-pending blueprint platform (Node.js/TypeScript, React, REST APIs) with rules-based recommendation engine, real-time data rendering, auto-certification workflows, and reusable software components across engineering teams.
 - **Requirements Analysis & Technical Design:** Work directly with architecture, infrastructure, product, security, and risk stakeholders to analyze requirements, design scalable workflows, define technical specs, and drive projects through full engineering lifecycle.
 - **Production Ownership & Level-2 Support:** Owned operational readiness for critical platform components through CI/CD, monitoring, runbooks, ADRs, incident triage, stakeholder communication, and escalation paths across regulated production environments.
 - **Team Leadership & Global Standards:** Led 5 engineers through sprint planning, code reviews, testing expectations, design discussions, and reusable architecture patterns adopted by 20+ teams across cloud, logging, and governance workflows.
- **HealthCard (Acquired Startup)** Remote
Blockchain Architect & First Full Stack Engineer Aug 2021 – Jul 2022
 - **Data-Driven Insurance Platform:** Joined as first engineer and built React/TypeScript frontend, Node.js/Express APIs, AWS EKS deployment, audit logging, and verification workflows; designed trust-score logic for real-time premium validation and fraud-resistant claims processing.
 - **ML/Data Pipeline Integration:** Integrated transformer-based document classification and monitoring for medical/insurance records, reducing manual review by 60%; built production APIs and feedback loops for credential verification.
- **Canada Revenue Agency (CRA)** Ottawa, Canada (Remote)
Software Developer Co-op Apr 2022 – Nov 2022
 - **Production Software Modernization:** Migrated federal tax platform UI from Apache Struts to React/TypeScript and contributed to backend APIs serving millions of Canadians with accessibility, quality gates, secure delivery, and maintainable code standards.
- **Dalhousie University** Halifax, Canada
Research & Teaching Assistant Jan 2021 – Aug 2022
 - **Algorithms, Systems & Cloud:** Researched call-stack decision algorithms and Python tracing tools for compiler/runtime analysis; TA for Algorithms, Software Development, and graduate Advanced Cloud Computing.

PROJECTS

- **Automatic Trading Algorithm:** Sub-second latency personal project integrating live brokerage APIs and real-time market data streams; momentum/mean-reversion signals with automated risk management, position sizing, and performance monitoring.
- **Portfolio Risk Optimization Framework:** Python research framework for market data ingestion, portfolio allocation, backtesting, Sharpe analysis, Monte Carlo simulation, risk-adjusted signals, and reproducible quantitative experiments.
- **Flight Price Optimization Agent:** DQN/DDQN reinforcement-learning agent in Python for dynamic pricing; modelled demand curves on historical flight data and improved simulated outcome by 18% over baseline.

EDUCATION

- **Dalhousie University** Halifax, Canada
MASc Applied Computer Science, GPA: 3.9/4.0 | TA: Algorithms, Advanced Cloud Computing Jan 2021 – Aug 2022
- **University of Toronto** Toronto, Canada
Enterprise Architecture Certification (3-semester program) Completed 2024
- **Gujarat Technological University** Ahmedabad, India
B.E. Computer Engineering, GPA: 3.92/4.0 Aug 2016 – Aug 2020